

The empirical recurrence for OEIS A266613, proved from an exact model

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Abstract

OEIS A266613 carries an empirical recurrence of order 3 contributed by Colin Barker. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by a proved period for the automaton's middle column, from which a provably satisfies a MONIC linear recurrence of order $S = 14$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A266613 is “Decimal representation of the middle column of the "Rule 41" elementary cellular automaton starting with a single ON (black) cell.”. It has offset 0 and begins

1, 2, 5, 10, 20, 41, 82, 165, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Conjectures from _Colin Barker_, Jan 02 2016 and Apr 16 2019: (Start) $a(n) = 2*a(n-1)+a(n-2)-2*a(n-3)$ for $n>5$. - [corrected by _Karl V. Keller, Jr._, Oct 07 2021]

As of the “Last modified” line on the live entry (Feb 16 2025 08:33:28, revision 63) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The middle column has a proved period

The middle column is the cell at the origin at each step, $c(n) = w(n)[n]$, the centre of the light cone; the entry lists the first $n + 1$ of them concatenated and read as a binary numeral. Nothing is being counted, so there is no digraph, and that the column settles into a period is an observation rather than a proof. The row certificate the automaton already carries gives one.

Write it as $w(n+p) = L_r + w(n) + R_r$ for $n \geq n_0$, $r = n \bmod p$. In absolute coordinates, where the row covers $x \in [-n, n]$ and $w(n)[i]$ is the cell at $x = i - n$, this says

$$\text{cell}(x, n+p) = \text{cell}(x + d_r, n), \quad d_r = p - |L_r|,$$

so the column reads a diagonal that moves by d_r every p steps. Three cases close it, and this automaton meets one of them:

- every L_r empty — the row grows only on the right, every prefix is frozen, so there is one infinite word W with $w(n)[i] = W[i]$ and $c(n) = W[n]$ outright; W is $w(n_0)$ followed by $R_{r_0}R_{r_1}\cdots$, eventually periodic with period dividing $\sum_r |R_r|$;
- every R_r empty — the mirror, with the word read from the right end;
- $\sum_r d_r = 0$ — the diagonal returns to the origin after a full cycle of residues, so $\text{cell}(0, n + p^2) = \text{cell}(0, n)$.

For this entry the column is periodic with period $q = 2$ from $n = 4$: the period above is a proved bound and the exact one is a divisor of it, found inside one window of computed rows, which is a finite check.

3 The bound

Concatenating gives $a(n + q) = a(n)B^q + v$ (the q new digits) with the same constant in each residue class, so a is annihilated by $(z^q - B^q)(z^q - 1)$. Allowing the pre-period to raise the numerator's degree gives a monic annihilator of order $S = 14$.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j < S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^3 - \sum_i c_i z^{3-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n - i)$ is a linear combination of shifts of a , so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a , hence every shift of a , hence any linear combination of shifts; that is r . A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. $\square$

Theorem 2.

$$a(n) = 2a(n-1) + a(n-2) - 2a(n-3)$$

for every $n > 5$.

Proof. The terms $a(0), \dots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 5+1$ onwards, and the run exceeds $S+3$, which by Lemma 1 settles every larger index at once. The bound is exact: at $n = 5$ the recurrence fails on the entry's own published terms, so no larger range holds. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 34 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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- [4] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Springer GTM 228, 2005.